Sarah Barnes Haddix

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Education

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA
B.S. Computer Science, B.S. Mathematics — 08/2022 - 05/2026
Graduated with Distinction and with Honors
Member, Honors Carolina (academic honors program admitting under 10% of each incoming class)
Dean's List: Fall 2022 – Spring 2026 (8 semesters)
Major GPA: 3.90/4.00 Overall GPA: 3.79/4.00

Research Interests

In April, I successfully defended my undergraduate thesis proving time-bounds and correctness for a proportional-share scheduling algorithm, advised by UNC Computer Science Department Chair Dr. James H. Anderson. After graduation, I will attend the Universität de Saarlandes to pursue a Master's degree in computer science. I have intentions of ultimately pursuing a PhD. My research interests lie broadly in real-time and distributed operating systems, robotics, and security. I am particularly drawn to theoretical and systems-level work at the intersection of these areas, including formal verification, concurrency, and the design of efficient and secure mechanisms for embedded or resource-constrained systems.

Professional Experience

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA
Undergraduate Research Assistant — 04/2024 - Present

- Working with Dr. James H. Anderson's Real-Time Systems Research Group, focusing on real-time operating systems, AI, and GPU scheduling
- Assisting graduate students in conducting experiments, for example exploring GPU memory constraints and overload behaviors, and providing feedback and proofreading on research papers
- Co-authored a real-time robotics paper accepted at ICRA25; major contributions included the development and modification of software in Python and C++ to control a simulated drone leveraging Linux and ROS and the optimization of a path planning neural network using Tensorflow and ONNX

Owl Cyber Defense — Morrisville, North Carolina, USA
Engineering Intern — 06/2025 - Present

- Working in Rust and Java to develop cross-domain network security solutions for US government applications
- Adding secure chat capabilities to existing products including the V2CDS, an NCDSMO Baseline Certified technology (Q3CY21), utilizing Apache Nifi, streaming XML data, and open source chat products like OpenFire
- Developing a secure RDP client/server in Rust for use in classified environments

Axai Tech — Cape Town, Western Cape, South Africa
Project Intern — 06/2024

- Leveraged deep learning techniques to produce new insights in the field of minimally invasive cancer treatment
- Engineered multiple deep learning models including an LSTM using PyTorch and a transformer model using HuggingFace libraries in order to classify and generate proteins according to project guidelines

Scale AI — Remote
Software Engineer for AI Training Data — 05/2024 - Present

- Contributing to sets of coding-related AI training data in mainly Java, Python, and Swift used to finetune modern LLMs

Lenovo — Raleigh, North Carolina, USA
Intern — 05/2021 - 06/2021

- Collaborated with a team of three other interns to compile and present research on emerging technologies (NFTs, cryptocurrencies) to a Lenovo executive

Publications

- **"On the Necessity of Real-Time Principles in GPU-Driven Autonomous Robots,"** Syed W. Ali, Angelos Angelopoulos, Denver Massey, Sarah Haddix, Alexander Georgiev, Joseph Goh, Rohan Wagle, Prakash Sarathy, James H. Anderson, Ron Alterovitz, *in Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*, March 2025

Activities & Projects

- **A Proportional Share Resource Allocation Algorithm For Real-Time, Time-Shared Systems: A Closer Look**; honors thesis proving time bounds and correctness for the earliest eligible virtual deadline first (EEVDF) scheduling algorithm and analyzing its implementation in the Linux kernel, passed thesis defense with honors
- **President of The UNC Chapter of the Association of Women in Mathematics**; organize and publicize events such as study nights, alumni/faculty panels, and social events to help foster a sense of community for women in the mathematics department; club is advised by Dr. Caroline Moosmueller
- **Member of UNC AI Club**; Developing a computer vision model for real-time wildfire detection optimized with TensorRT to be deployed on a Jetson-powered drone. Implementing a novel messaging protocol to stream live fire detection data from the drone's camera to a remote server.
- **Selected for UNC's Undergraduate CyberSecurity Engagement Joint Program with The University of Nagoya in Japan**; exchange program including a trip to Nagoya, Japan to explore the intersection of cybersecurity, national security, and foreign policy through workshops, lectures, and site visits with industry and government leaders
- **Formal Verification of Onion Franking Security Properties Using Tamarin**; semester long project in COMP 590/637 to formally verify certain properties of a metadata-hiding cryptographic protocol that enables anonymous abuse reporting, proposed by researchers at The University of North Carolina in *Onion Franking: Abuse Reports for Mix-Based Private Messaging*, NDSS 2025
- **Awarded "Best Functionality" in Carolina App Team's Showcase, Spring 2024**, for a fully functional recreation of the Shazam app using ShazamKit API; completed App Team Carolina's 2 semester-long iOS development program, building several functional iOS apps